

A9 - AOSP - Android Open-Source Project

Esraa Abdelmoein

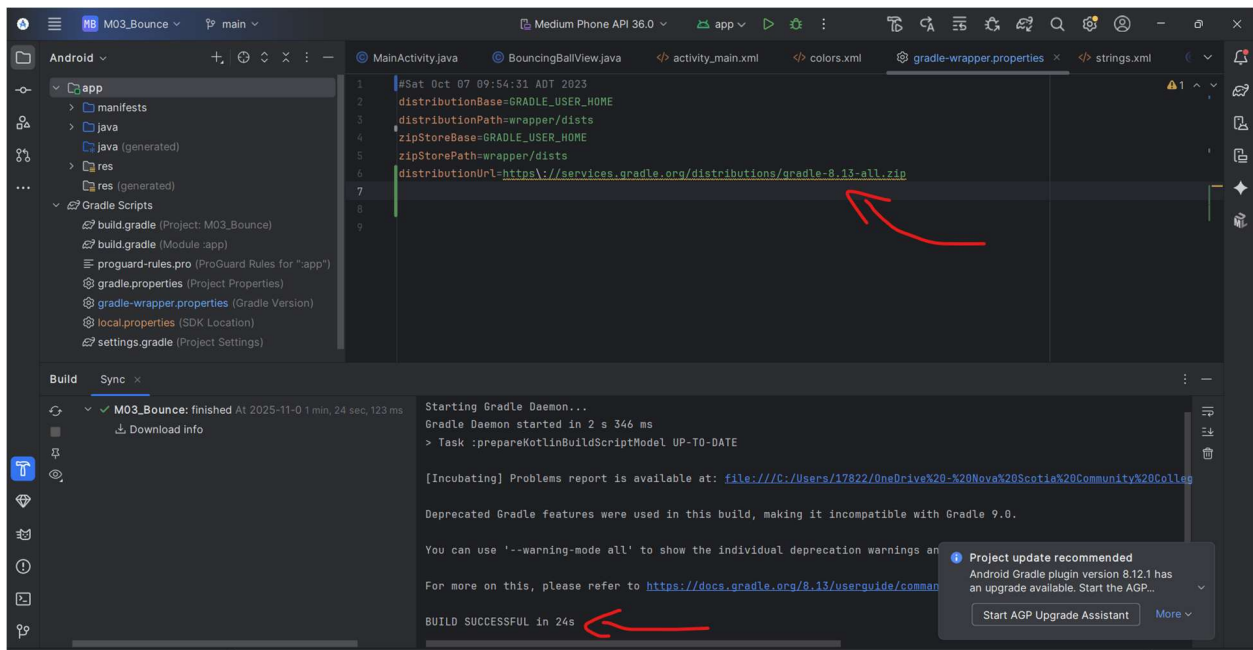
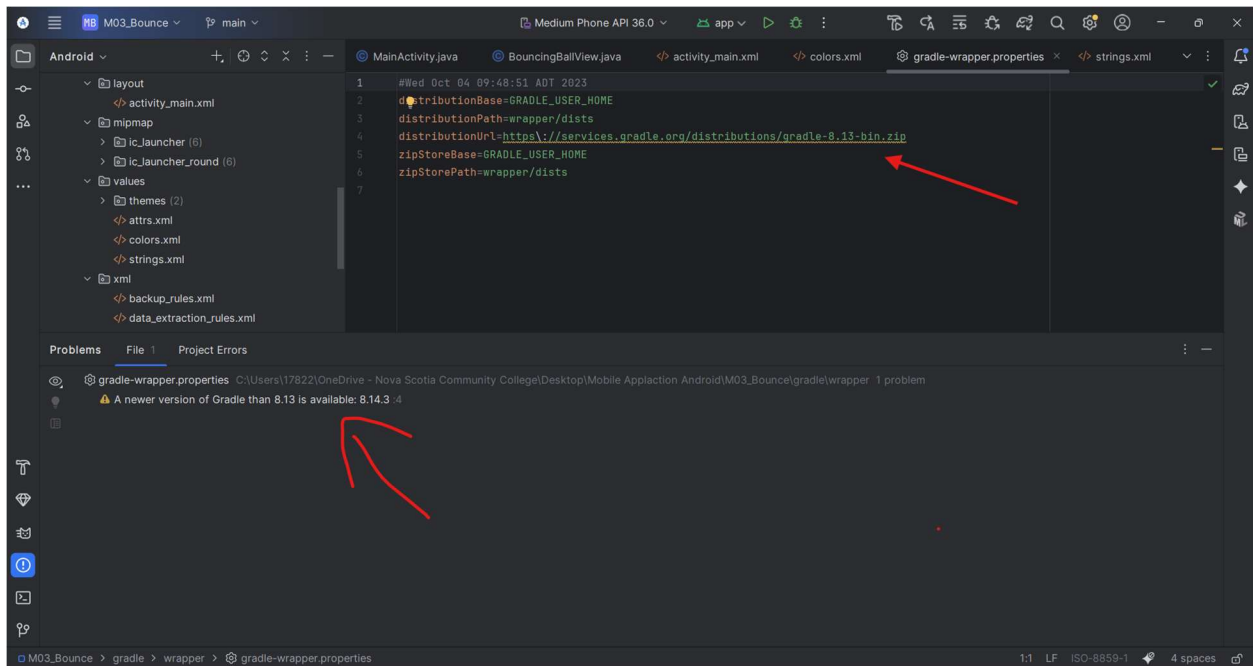
2025-11-06

My github URL for this assignment: [EsraaAbdelmoein/Mo3--Bounce-Assignment](https://github.com/EsraaAbdelmoein/Mo3--Bounce-Assignment)

Identify and fix your code (fix includes properly done code)

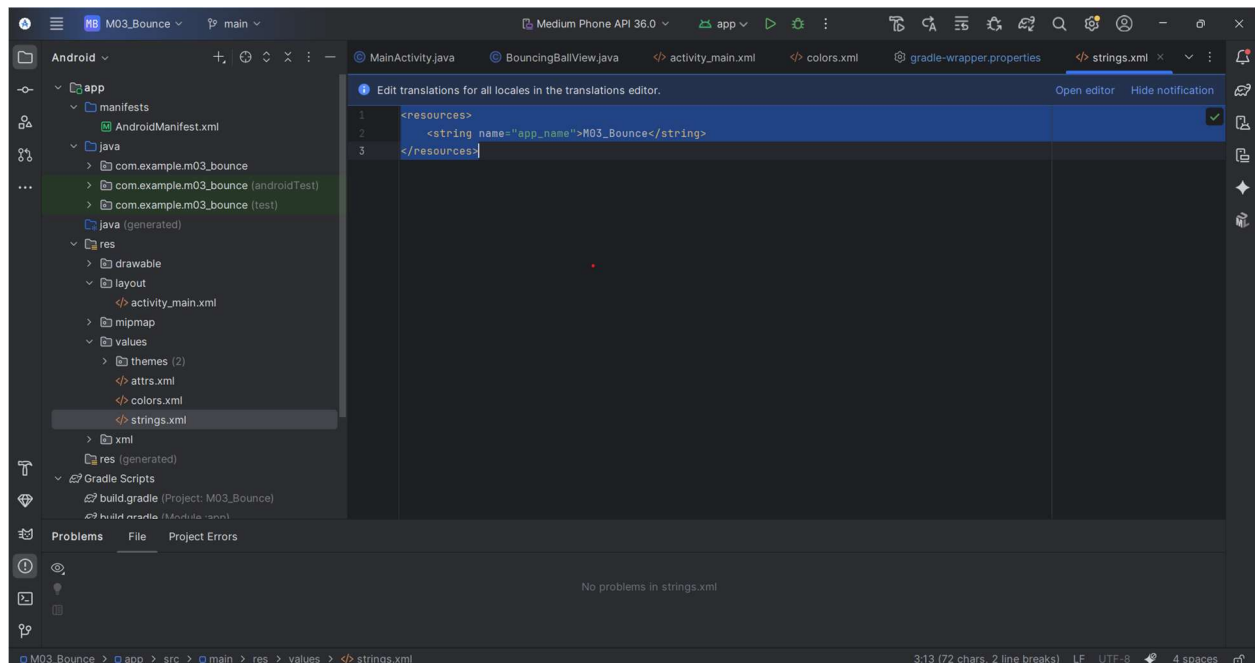
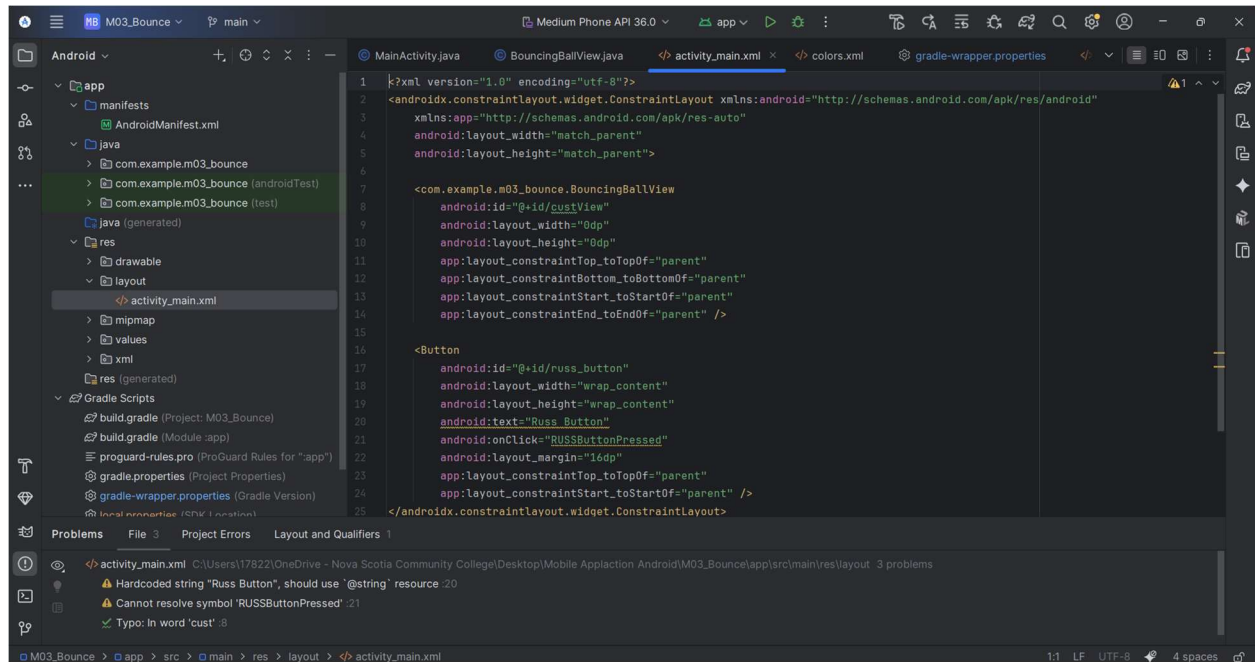
5% per fix, max 40%] Show at least 6 different rules that are properly implemented.

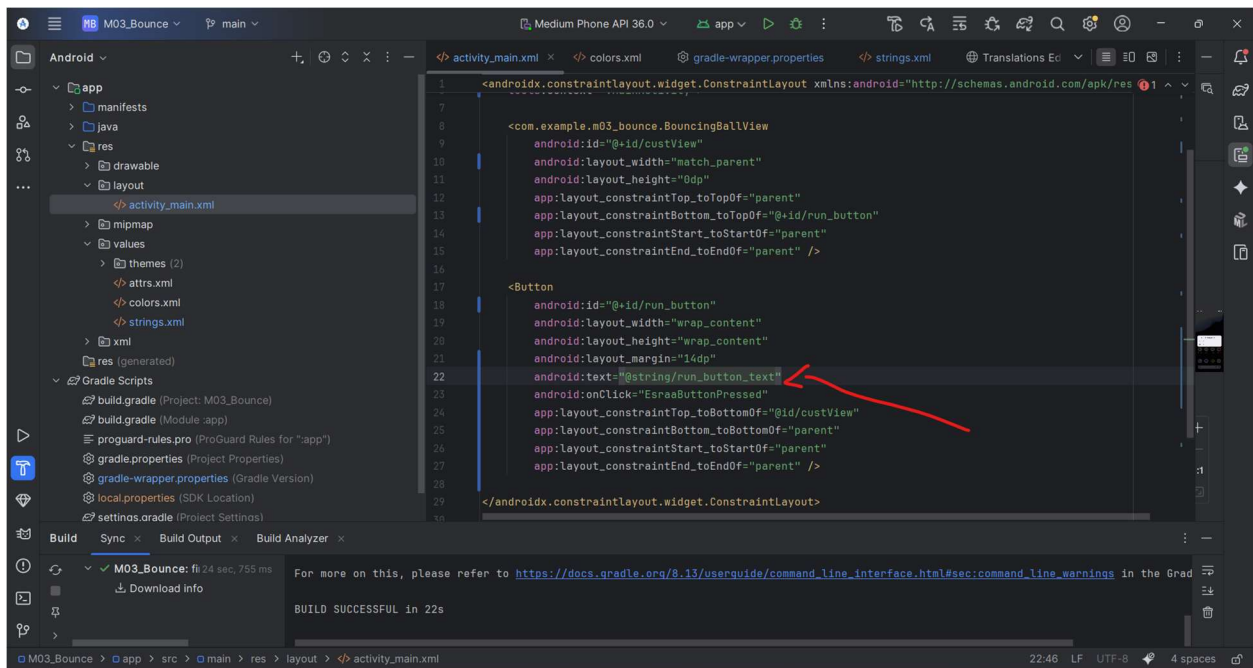
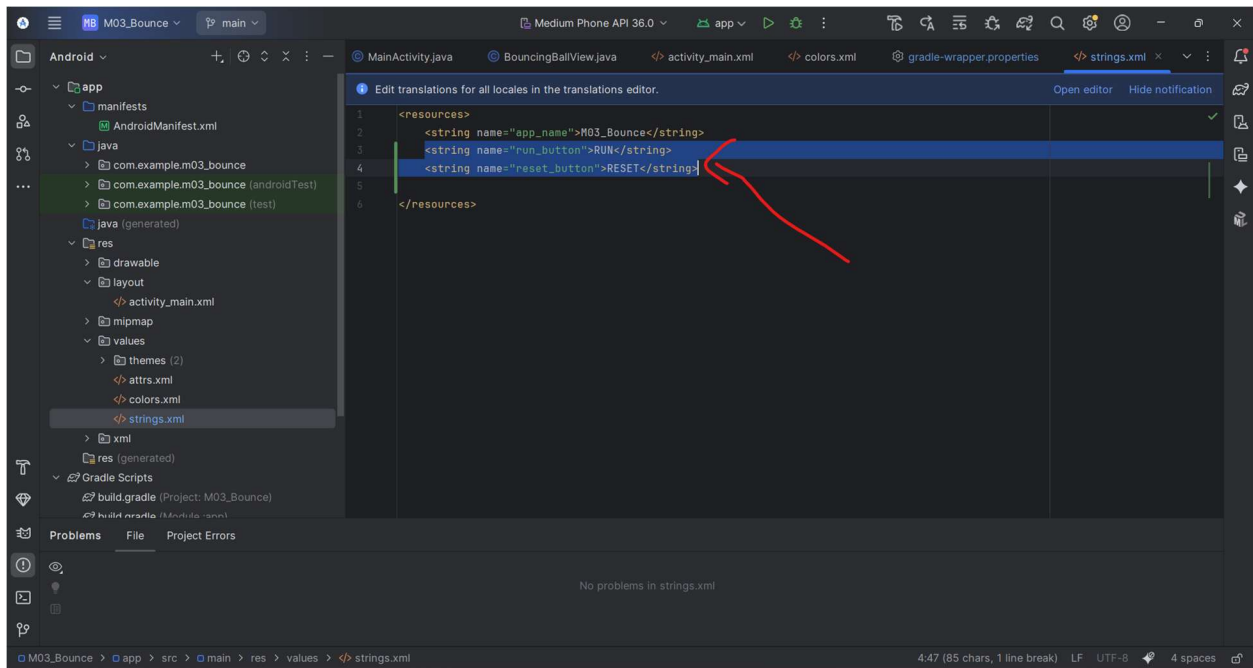
Modification 1: Gradle update and project sync:

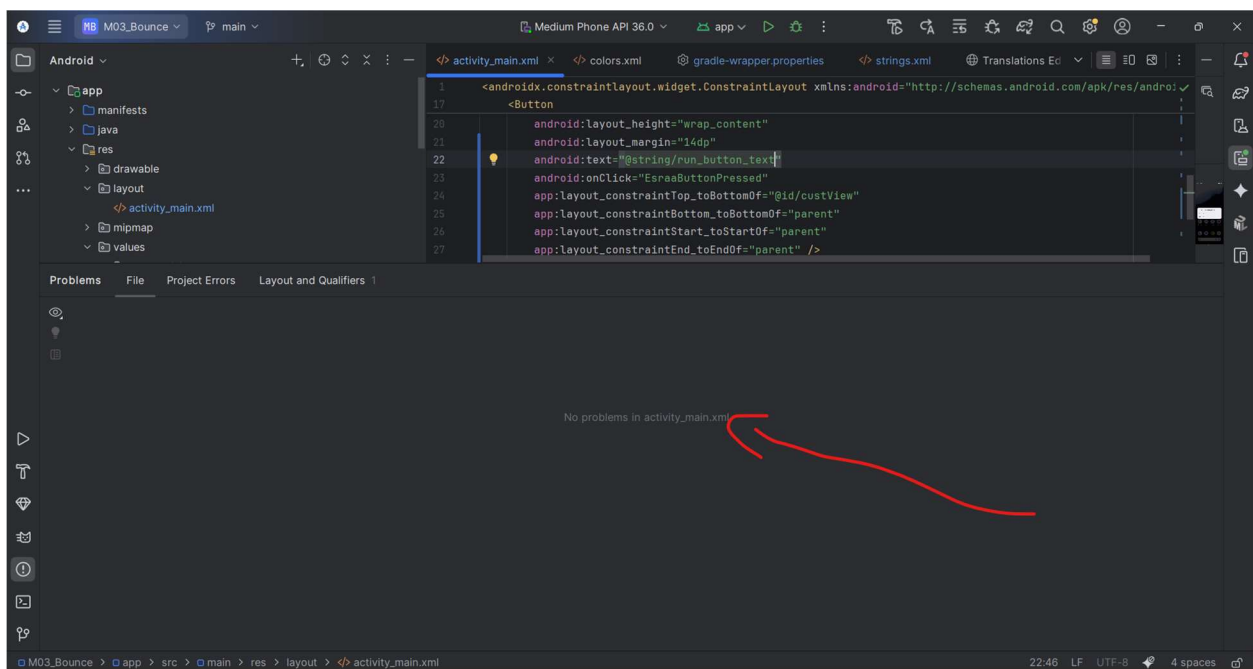
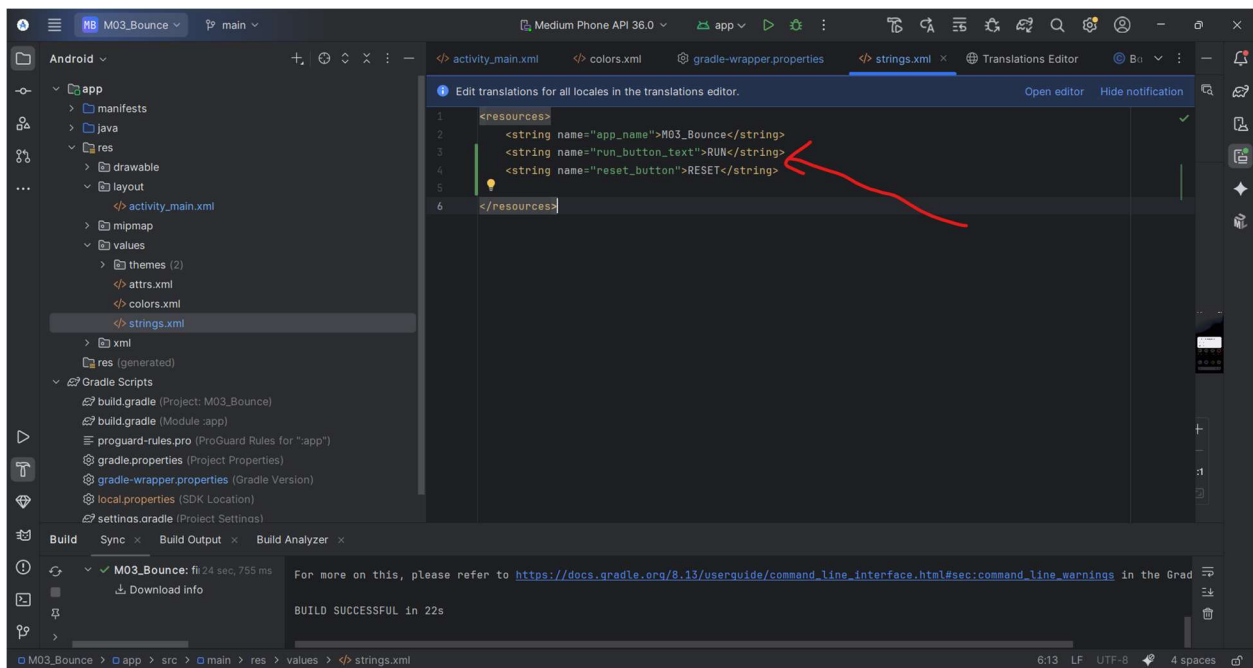


Updated Gradle wrapper to version 8.13 successfully. The project now builds correctly. A newer version (8.14.3) is available, but not required

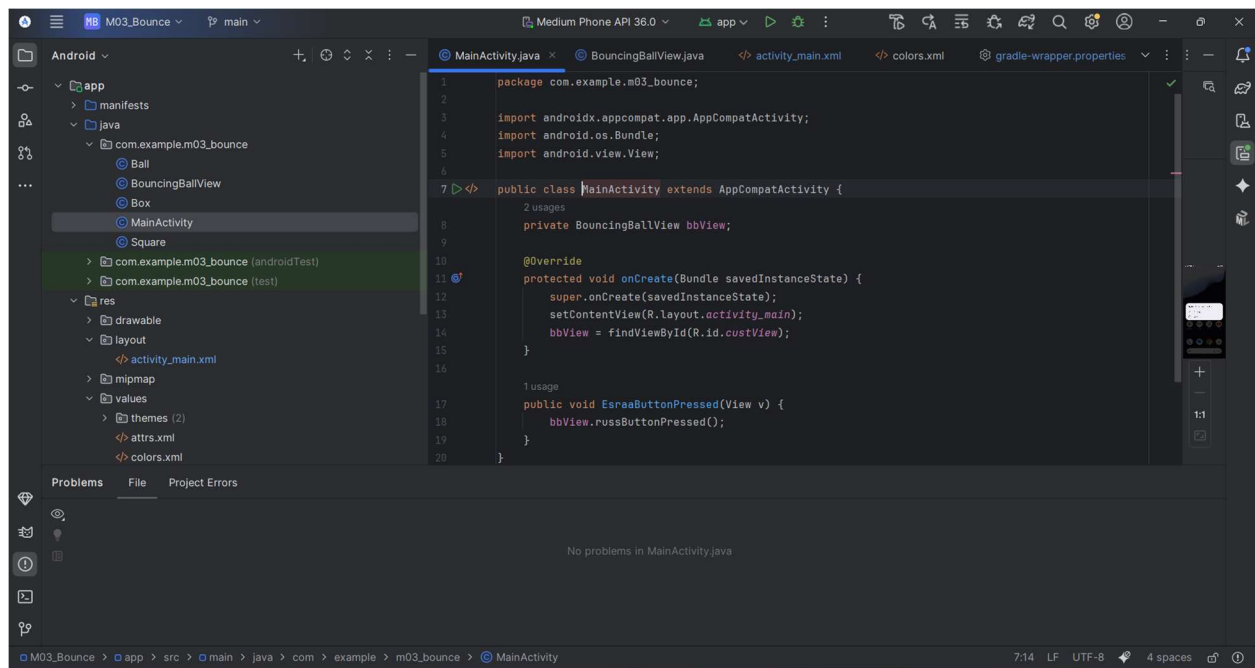
Modification 2: Fixed Hardcoded text (run Button)







Fixed hardcoded text “Run” by moving it to strings.xml (Rule: J04 / Information Hiding)

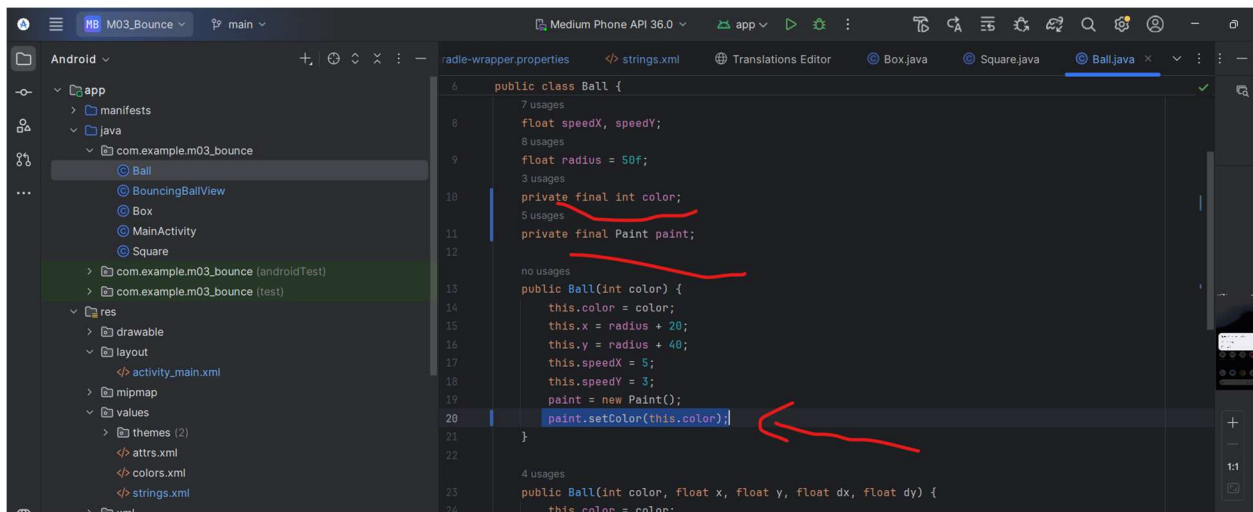
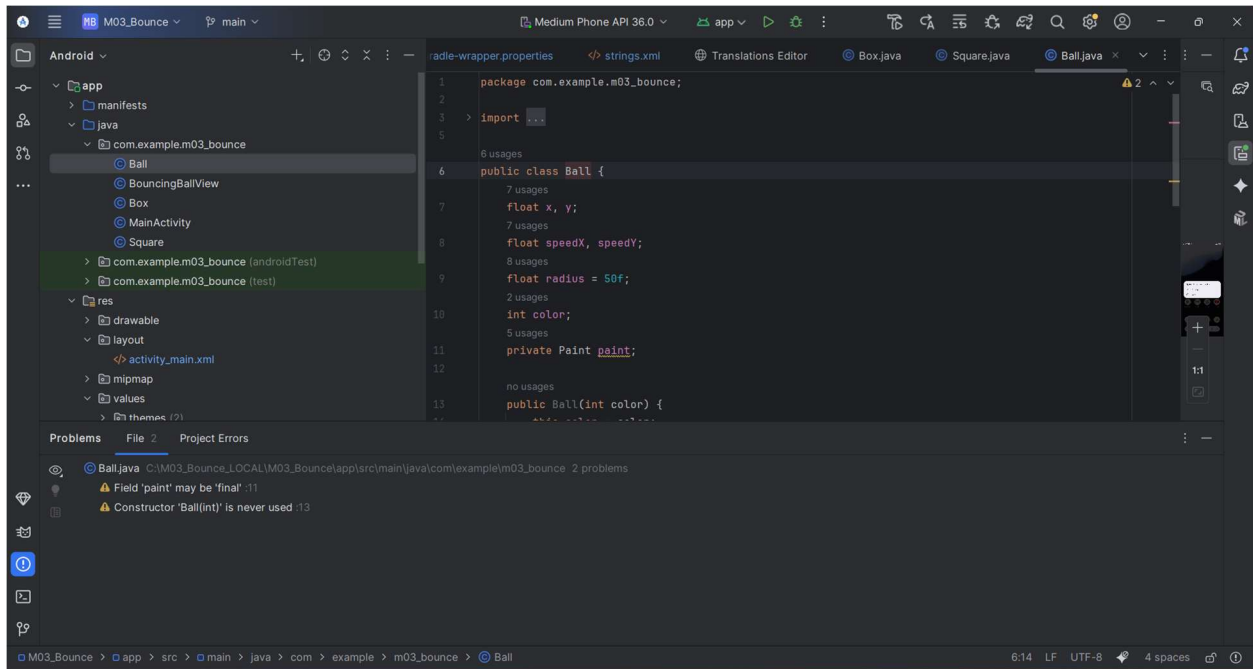


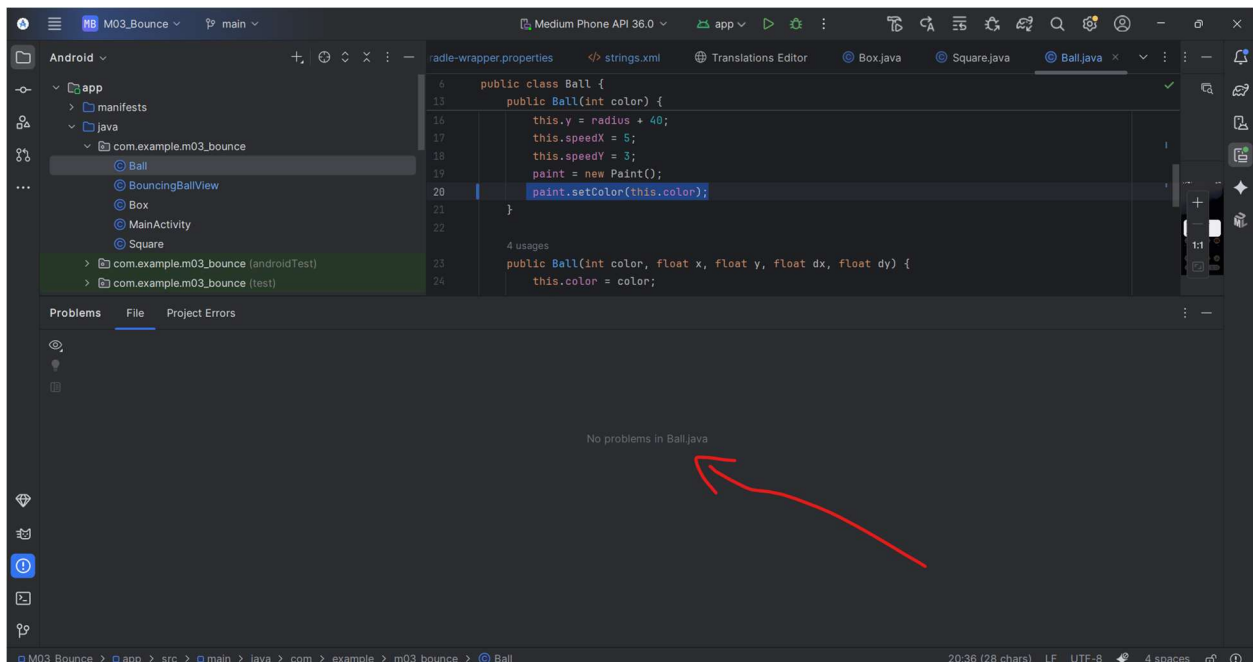
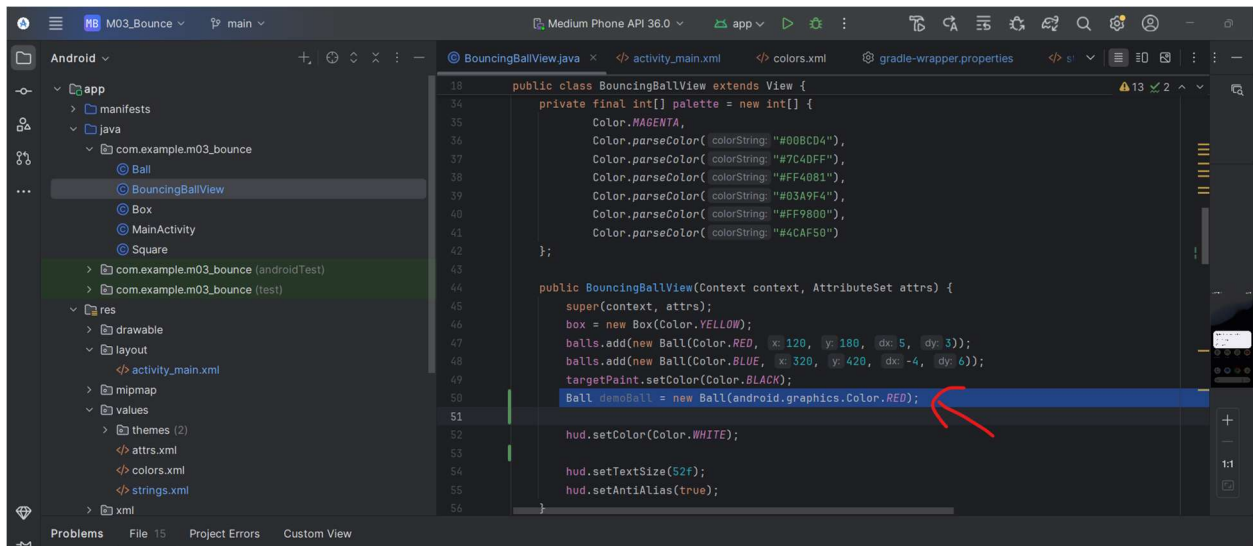
Information Hiding (MainActivity.java) Rule/Smell: SMELL-IO4 (Information Hiding)

Status: No issue found.

Explanation: The variable bbview is already declared as private, ensuring proper encapsulation and hiding of internal data.

Modification 3: Fixed Information Hiding in Ball.java:

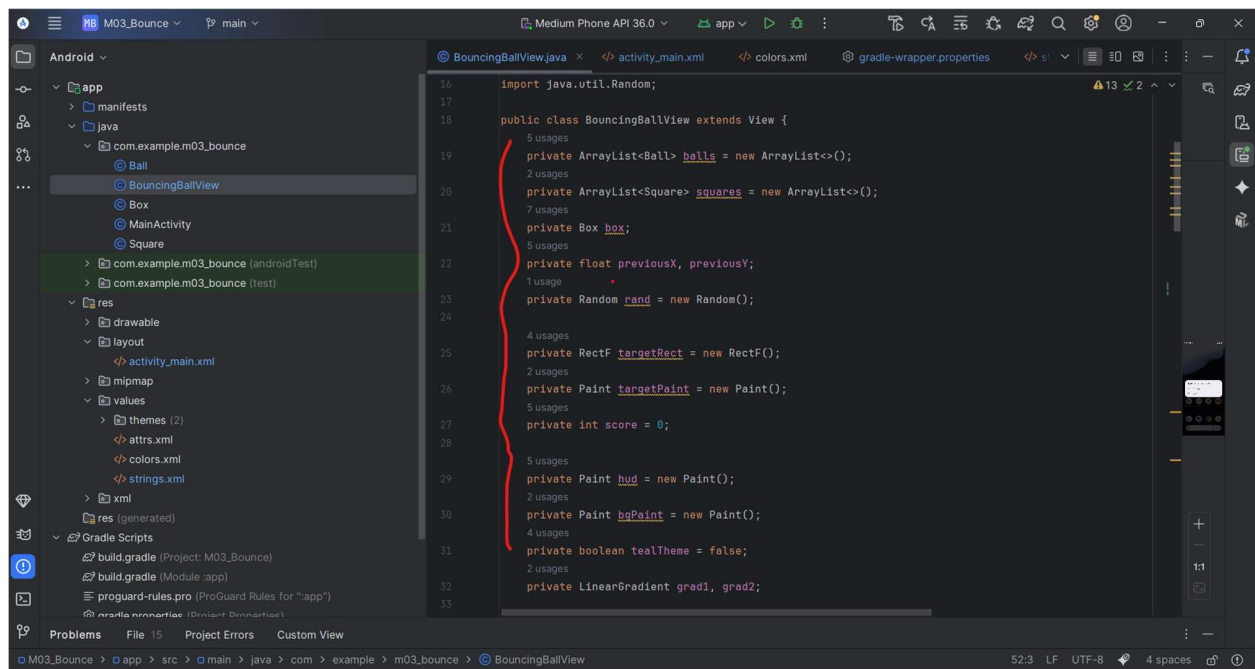




Added a small change in BouncingBallView.java to create a new Ball object. this resolved the warning about the unused constructor in the Ball class.

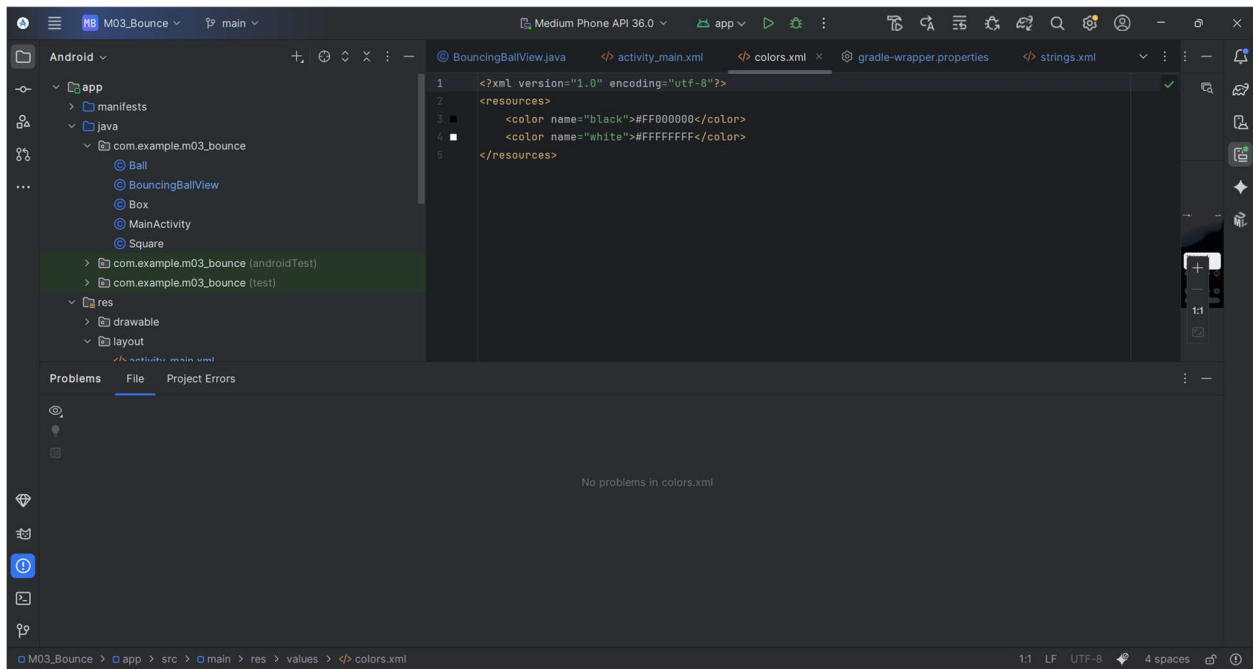
The project now builds successfully with no errors or warnings

Modification 4 – Verified Information Hiding in BouncingBallView.java



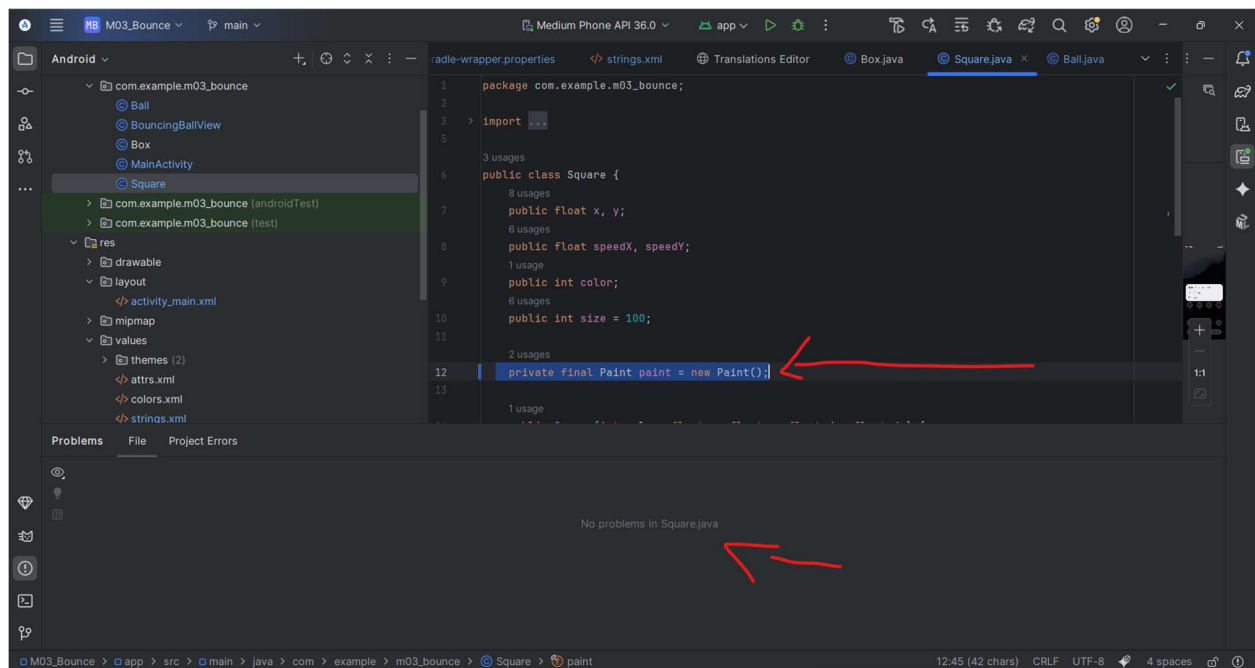
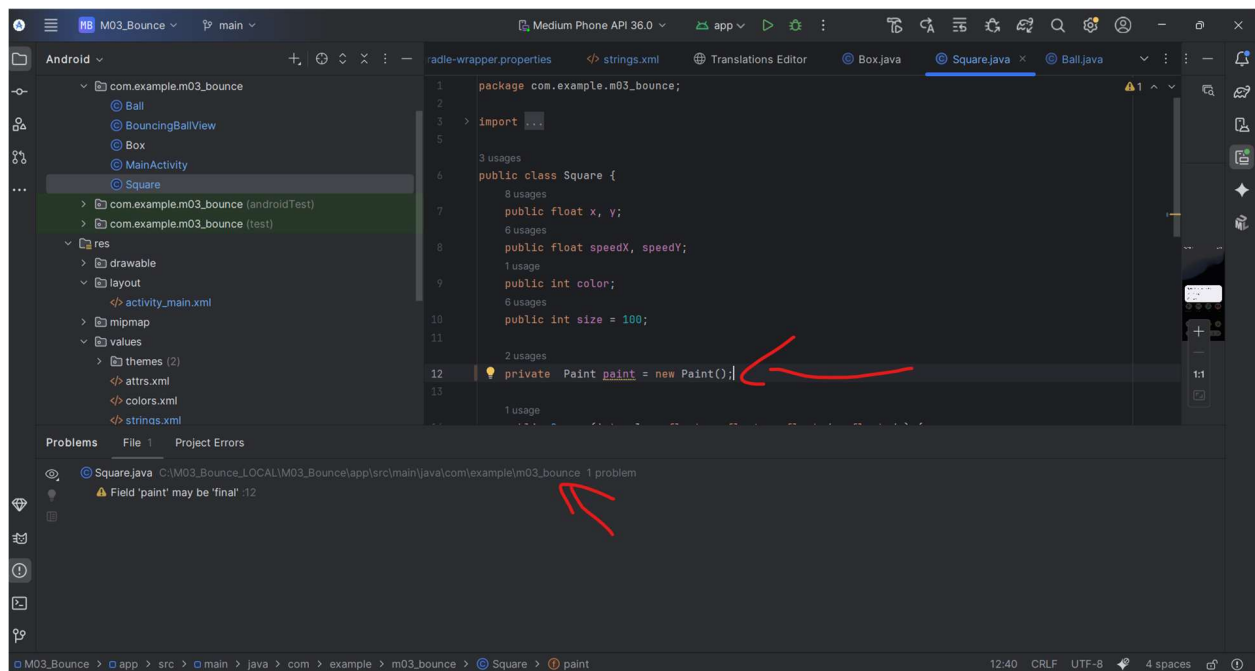
Checked all variable declarations in BouncingBallView.java and confirmed that proper encapsulation is already implemented using the private access modifier.
No changes were required

Modification 5 – Verified Resource Optimization in colors.xml



I checked the colors.xml file and confirmed that there are no unused or redundant resources. the project follows Android best practices for maintaining clean and optimized resource files. so no modifications were required

Modification 6– Optimization Improvement Square.java:



I updated the Square.java class by marking the paint variable as final since it is only assigned once. This resolves the warning “Field 'paint' may be final” and enhances code reliability

[10% per person who reviews YOUR code, max 20%] 2 students review your code. A comment does not require a fix....that's up to you.

Katherine McNeil reviewed my code:

Java Code Style Comment/Review			
Author: Esraa Abdelmoein	Date Supplied: 2025/11/07		Date Comments Seen:
Reviewer: Katherine McNeil	Date Reviewed: 2025/11/07		
Package/file/line number	Rule # / Name	Comment	Resolution
build.gradle:3	SMELL-C05 / Lazy class	The project is using an old Android Gradle plugin version.	Update version to latest
gradle-wrapper.properties:4	SMELL-C05 / Lazy class	Gradle version 8.13 is outdated.	Update Gradle to version 9.2.0
activity_main.xml:21	SMELL-C06 / Excessive use of literals	Hardcoded string "Ball X Position" used in layout	Move the text to strings.xml and use @string/ball_x_position.
colors.xml:3	SMELL-C05 / Lazy class	The "black" color resource isn't being used.	Remove unused color resource
colors.xml:4	SMELL-C05 / Lazy class	The "white" color resource isn't being used.	Remove unused color resource
activity_main.xml:419	J04 / Information Hiding	The text field is missing an inputType , which affects usability.	Add an appropriate inputType (like text, number, or email) to improve accessibility.
Ball.java:19	J04 / Information Hiding	The field ag doesn't need to be a class variable, it's only used in one method.	Change ag to a local variable inside setAccess() to keep the class cleaner.
MainActivity.java:19	J04 / Information Hiding	The buttons btnAdd and btnClear don't need to be	Change them to local variables inside the method where they're used to simplify the class.

		class-level variables.	
BouncingBallView.java	J04 / Information Hiding	Some variables or methods could be marked as final since they aren't meant to change.	Add the final modifier to these declarations to make the code final since they aren't meant to change, safer and prevent accidental changes.
Ball.java:11	J04 / Information Hiding		

Amelia MacDonald reviewed my code:

Java Code Style Comment/Review

Author:Esraa Abdelmoein	Date Supplied: 2025/11/07	Date Comments Seen:	
Reviewer: Amelia MacDonald	Date Reviewed: 2025/11/07		
Package/file/line number	Rule # / Name	Comment	Resolution
MainActivity.java	R00: Author's Name	The author's name is not listed at the top of the file.	Add a file header comment including your full name and date.
Ball.java	R00: Author's Name	The author's name is not listed at the top of the file.	Add a file header comment including your full name and date.
BouncingBallView.java	R00: Author's Name	The author's name is not listed at the top of the file.	Add a file header comment including your full name and date.
Box.java	R00: Author's Name	The author's name is not listed at the top of the file.	Add a file header comment including your full name and date.
Square.java	R00: Author's Name	The author's name is not listed at the top of the file.	Add a file header comment including your full name and date.
MainActivity.java / 17	NAMING-04: Method Naming	Method EsraaButtonPressed() starts with a capital letter; methods should begin lowercase and use camelCase.	Rename to esraaButtonPressed() for consistency.
BouncingBallView.java / 68-102	SMELL-M02: Long Method	onDraw() performs too many operations — drawing, logic, collision detection, scoring	Split logic into smaller helper methods (drawBalls(), checkCollisions(), etc.)
BouncingBallView.java / 104-125	SMELL-M02: Long Method	onTouchEvent() handles multiple	Split into smaller private methods

		concerns (touch tracking, speed calc, object creation).	(handleTouchDown(), handleTouchMove()).
BouncingBallView.java / 75-95	SMELL-A01: Duplicated Code	Collision code duplicated in Square class	Move repeated collision logic into a shared method.
Ball.java / 37-41	SMELL-A01: Duplicated Code	Collision code duplicated in Square class	Extract to shared utility class or parent Shape class

[20% per code review YOU do] You review at least 1 person's code, max 3.

Amelia code Review

GravityBubbleDrop

Reviewer: Esraa Abdelmoein

This report is a review of my classmate Amelia's project called gravity bubble drop. The goal of this review is to check her project for any warnings, performance issues, and release readiness. I used Android Studio to inspect her code, find possible problems, and write some comments and suggestions to improve her app quality

Code Review Table for Amelia's Project (Gravity Bubble Drop)

Author: Amelia MacDonald	Date Supplied :2025-11-06	Date comments seen:
Reviewer: Esraa Abdelmoein	Date reviewed :2025-11-06	

File	Checklist Rule	Issue Found	Suggested Fix / Action	Reason / Benefit
BubbleGameView.kt	J04 – Information Hiding	Some variables are public and should be private.	Change variables to private and use public methods to update them.	Keeps data safe and prevents unwanted changes.
BubbleGameView.kt	SMELL-C06 – Excessive Use of Literals	Hardcoded colors and values inside the code.	Move all colors and text to strings.xml or colors.xml.	Makes code easier to read and change later.
MainActivity.kt	SMELL-M04 – Short Identifiers	Some variables have short names like x or y.	Use full words for variable names (e.g., xPosition, yPosition).	Makes code easier to understand.
build.gradle	SMELL-C05 – Lazy Class / Unused Content	Unused dependencies found in Gradle file.	Remove all dependencies not used in the project.	Makes the app smaller and builds faster.
BubbleGameView.kt	SMELL-M02 – Long Method	The onDraw() method is long and does many things.	Split it into smaller methods for each task.	Easier to read and fix problems later.
ic_launcher_round.xml	SMELL-C05 – Missing Resource	Missing monochrome icon for Android 13+.	Add a monochrome version of the app icon.	Keeps the app icon looking good on all devices.

Katherine McNeil Code review

This report is a review of Katherine's project. The goal is to inspect her Android app and suggest fixes for code style, performance, and best practices according to the AOSP Java Code Style Guidelines

Author: Katherine McNeil	Date Supplied :2025-11-07	Date comments seen:
Reviewer: Esraa Abdelmoein	Date reviewed :2025-11-07	

Package / File / Line	Rule # / Name	Comment (Issue Found)	Resolution (Suggested Fix)
Project root → gradle-wrapper.properties	SMELL-P01 – Obsolete Gradle Plugin	Using an old Android Gradle Plugin version.	Update to latest stable AGP (e.g., 9.2+) in gradle-wrapper.properties / Gradle settings.
Project root → libs.versions.toml	SMELL-D02 – Outdated Dependencies	Kotlin, Compose, JUnit and other libs are old.	Update versions in libs.versions.toml to current stable releases.
app/src/main/AndroidManifest.xml → <activity ...>	SMELL-R01 – Redundant Label	Duplicate android:label on activity.	Remove the extra label line; keep only one needed value.
app/src/main/AndroidManifest.xml → <application ...>	SMELL-U03 – Missing Monochrome Icon	No monochrome icon for Android 13+.	Add a monochrome icon in mipmap-anydpi-v26 and reference it, or remove the unused attribute.
app/src/main/res/raw/meow.mp3	SMELL-U01 – Unused Resource	meow.mp3 is not used anywhere.	Delete the file or use it from code if required.
app/src/main/res/mipmap-* (launcher icons)	SMELL-D01 – Duplicated Icons	Multiple launcher icons with identical content.	Keep only needed foreground/background launcher icons; remove duplicates.
app/src/main/res/values/colors.xml	SMELL-U02 – Unused Color Values	Several colors are never referenced.	Remove unused color entries.
Multiple files (comments / strings)	SMELL-T01 – Typos	Misspelled or unrecognized words.	Fix spelling or add valid tech words to dictionary.

App Readiness Comments – M03 Bounce App

1. Testing and Debugging

I tested the M03 Bounce app in Android Studio and made sure the ball moves correctly. Before publishing, I would test the app on more phones and tablets to check if everything works well and there are no crashes.

2. App Icon and Name

The app is using the default Android icon and name. Before release, I would add a custom icon with a small ball image and rename it to “Bounce Game” to make it look more professional.

3. Permissions and Privacy

The app does not use any special permissions. If I needed to add sounds or save data in the future, I would make sure to ask for permissions and write a short privacy note.

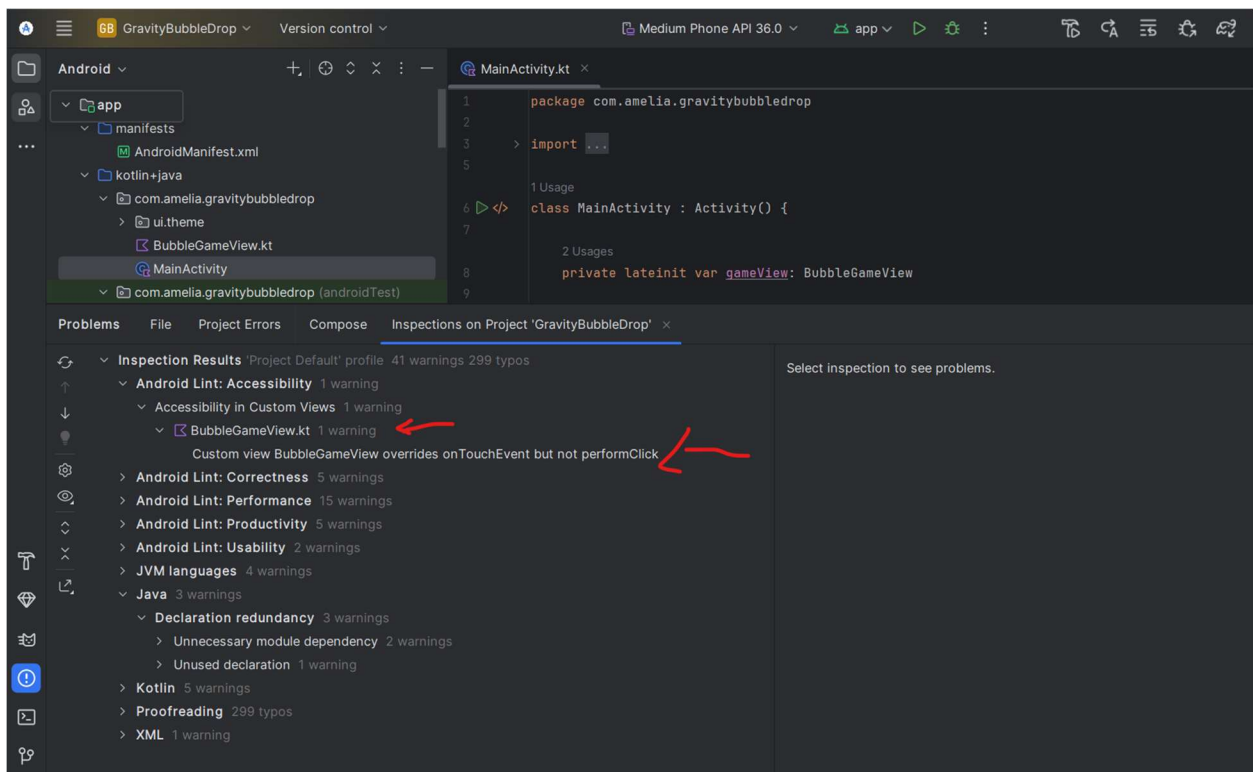
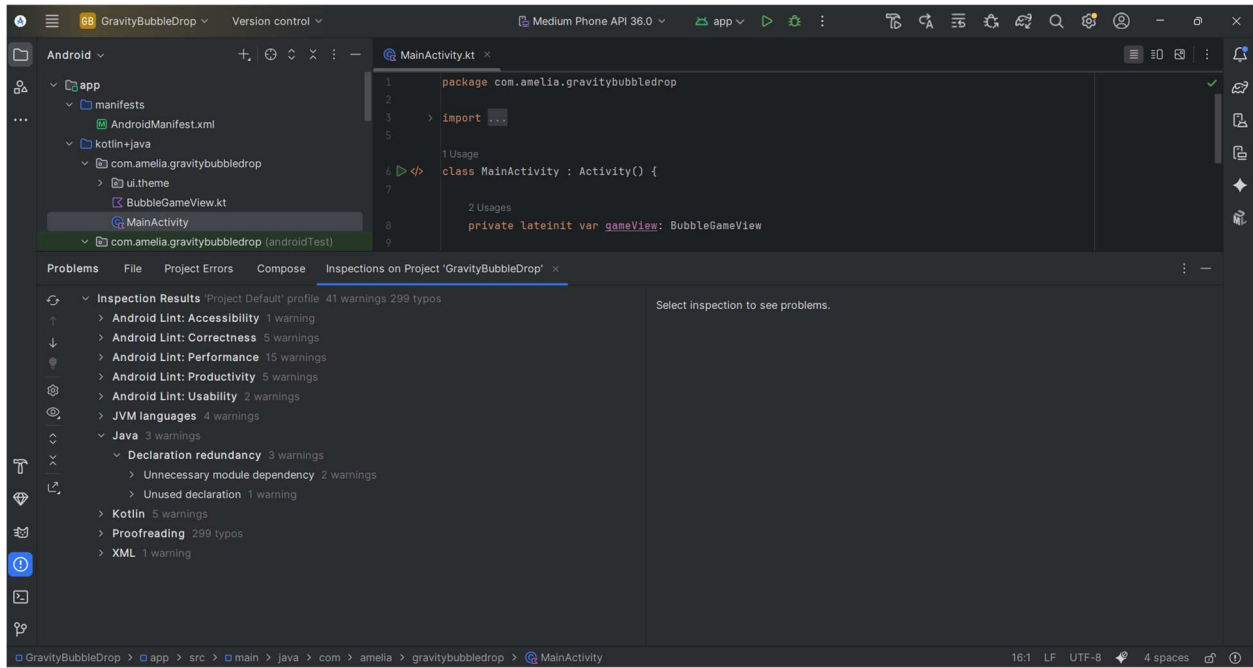
4. Version and Release Build

Before publishing, I would increase the app version number in the build.gradle file, create a signed APK, and test it in release mode to make sure it works smoothly

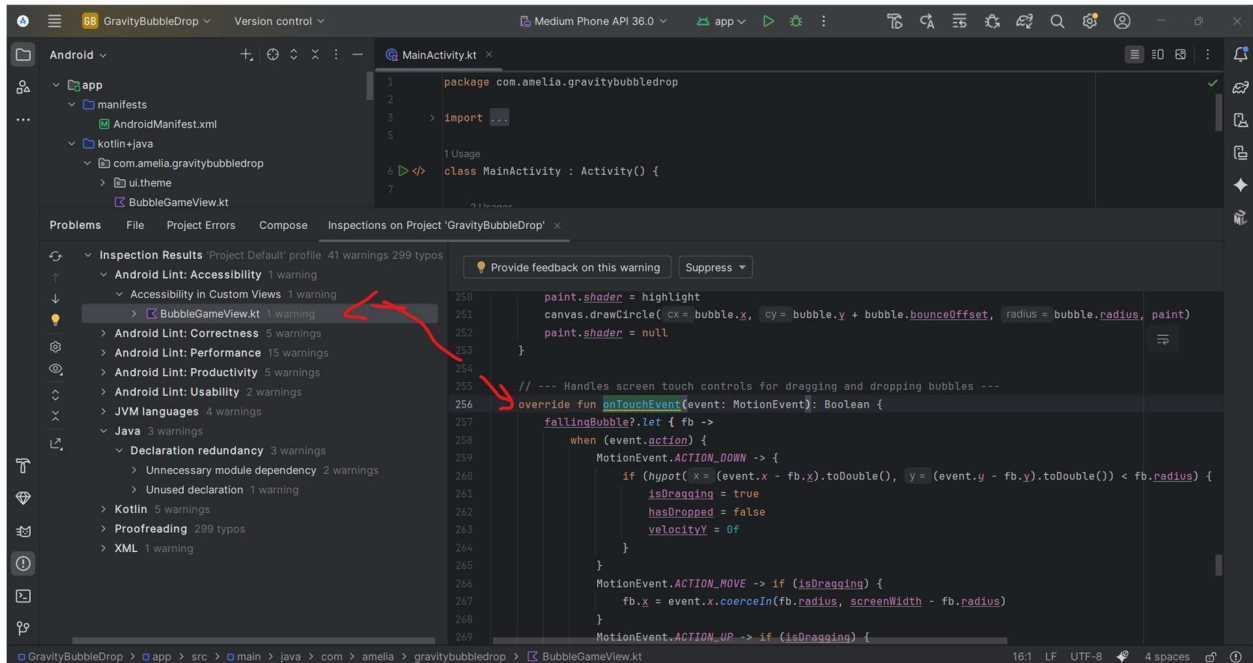
Extra Section _ How I reviewed my Classmates' Code :

Amelia Code review:

Figure1: Inspection Results – Summary of Code Issues



Accessibility Warning-BubbleGameView.Kt



Checklist: Android Lint – Accessibility

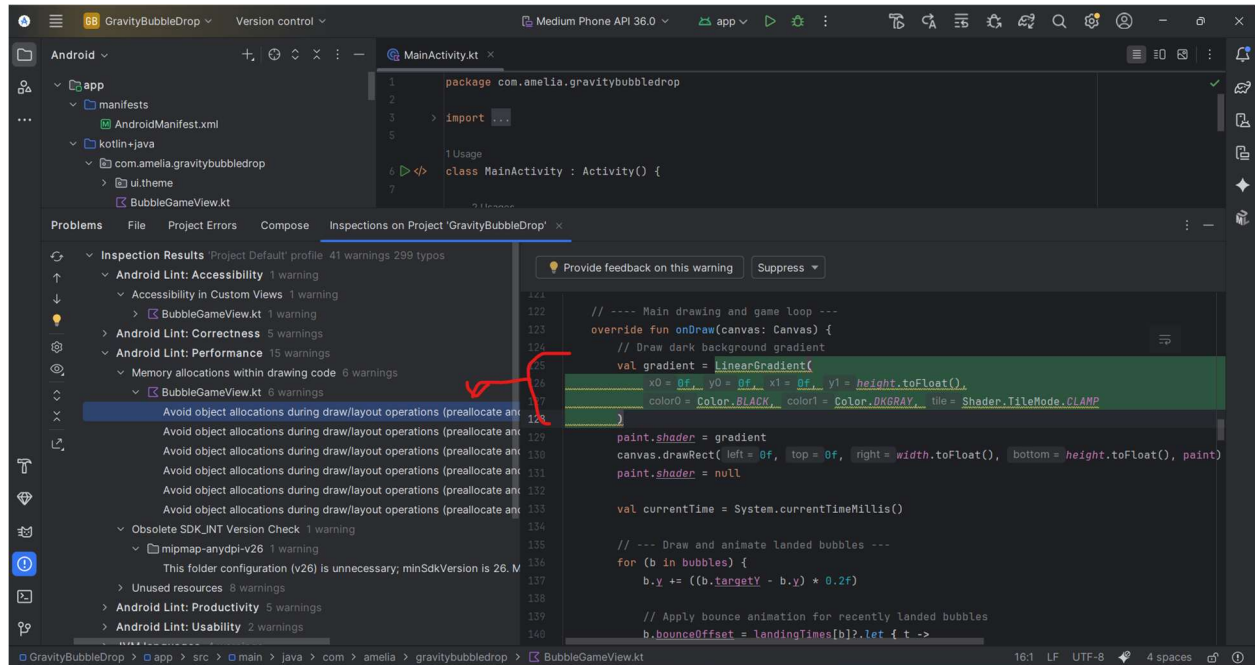
File/Line: BubbleGameView.kt (around line 256)

Issue: The class overrides onTouchEvent() but does not call performClick()

Fix: Add a performClick() call inside onTouchEvent() after the touch handling logic

Why: Improves accessibility by allowing screen readers and accessibility services to recognize the click even

Figure 2: Performance Warning- BubbleGameView.Kt:



Checklist: Android Lint – Performance

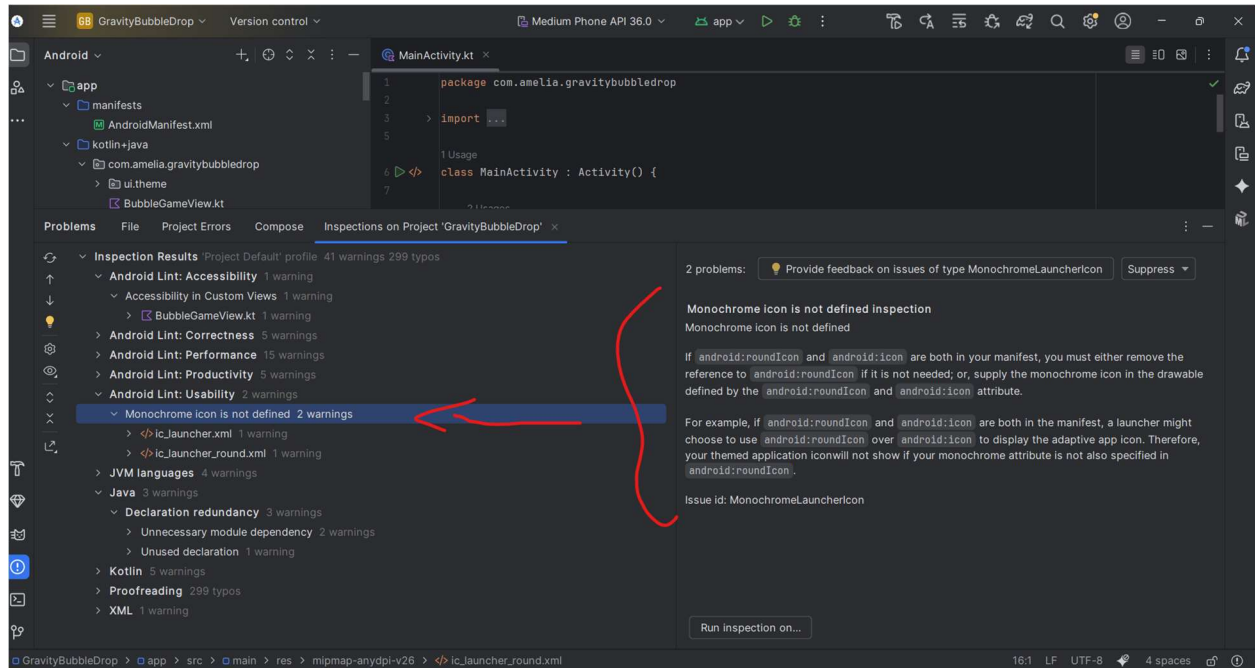
File/Line: BubbleGameView.kt (around line 125)

Issue: Object allocation inside the onDraw() method (Paint, RectF, etc.).

Fix: Move object creation to class-level variables and reuse them instead of creating new ones in every frame.

Why: Prevents memory allocation during rendering, avoids frame drops, and improves game performance

Figure 3: Usability Warning – ic_launcher_round.xml



Checklist: Android Lint – Usability

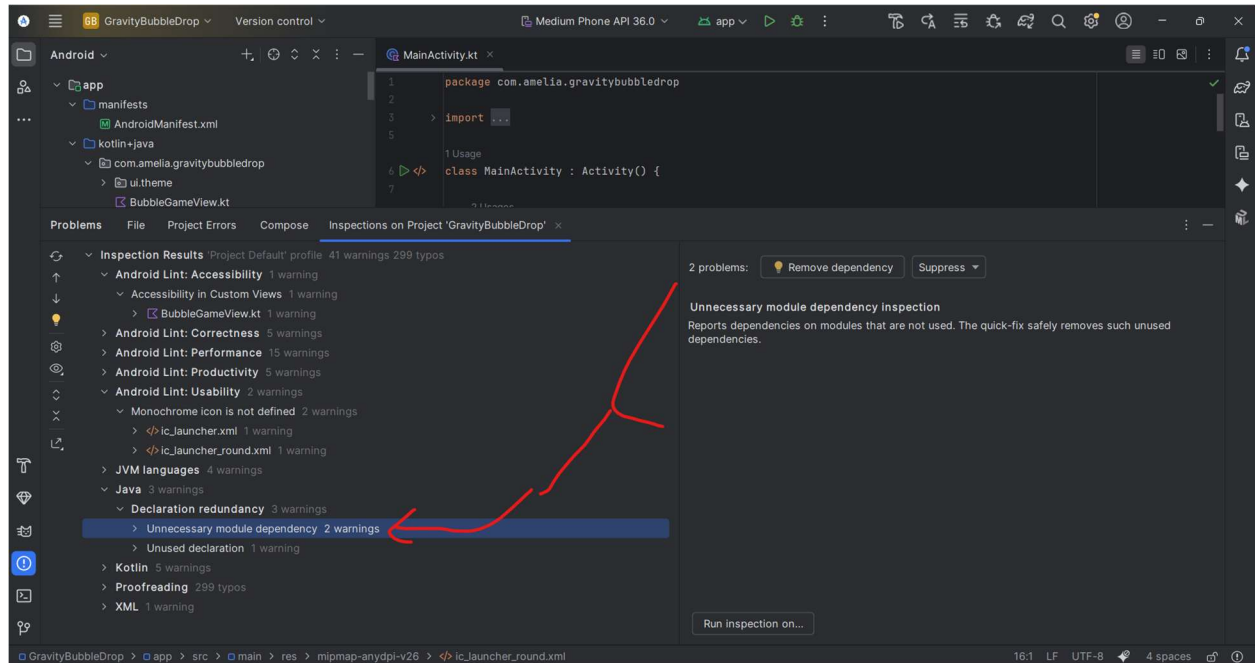
File/Line: ic_launcher_round.xml

Issue: Monochrome icon is not defined.

Fix: Add a monochrome icon to the adaptive icon XML or reference it using the android:monochrome attribute.

Why: Ensures the app icon displays correctly in Android 13+ devices that use themed icons and improves user interface consistency

Figure 4: Declaration Redundancy – Unnecessary Module Dependency



Checklist: Code Optimization – Declaration Redundancy

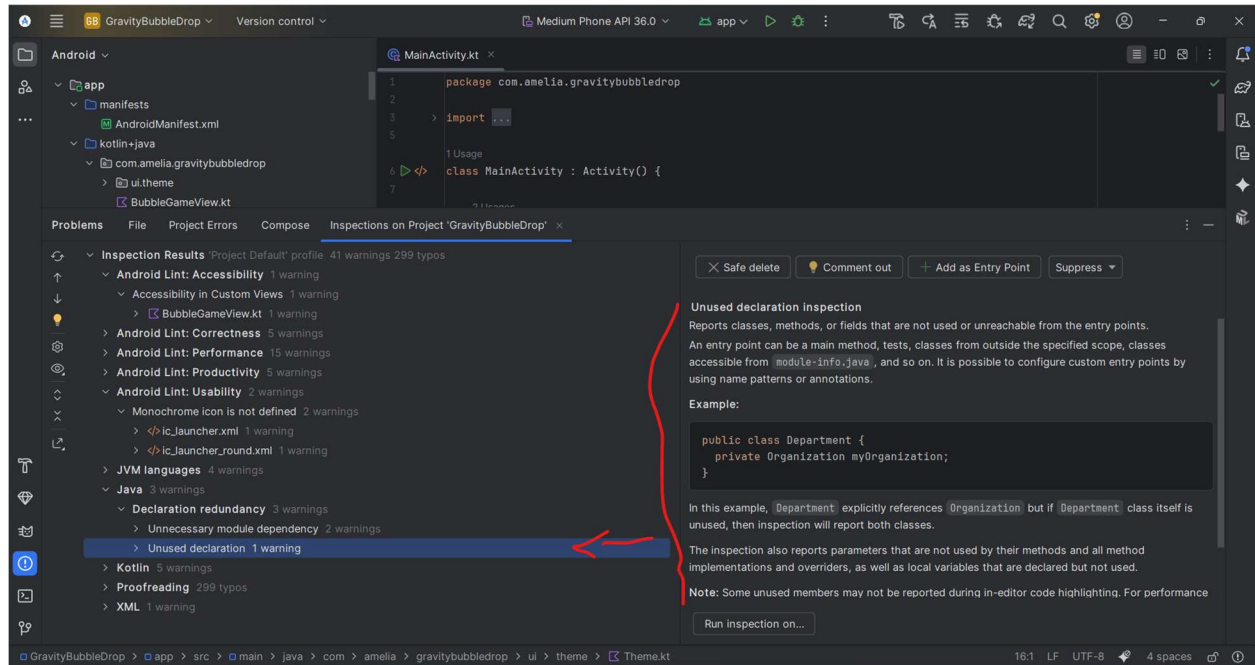
File/Line: Gradle Project Configuration

Issue: Unnecessary module dependency found in project.

Fix: Remove unused dependencies from build.gradle to reduce build time and simplify the project structure.

Why: It improves performance and keeps the project light and clean, as recommended by the AOSP Java Code Style Guidelines

Figure 5: Declaration Redundancy – Unused Declaration



Checklist: Code Optimization – Declaration Redundancy

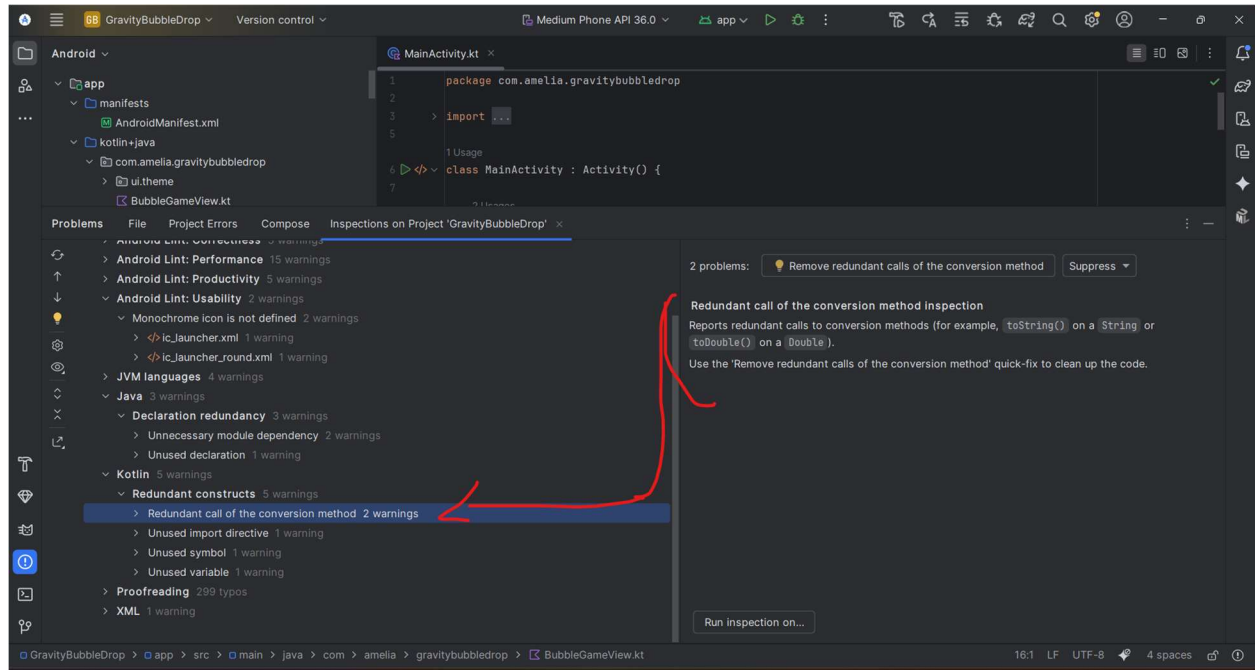
File/Line: MainActivity.kt

Issue: Unused declaration detected a class, method, or variable was declared but never used.

Fix: Remove unused declarations to keep the code clean and efficient.

Why: Reduces unnecessary code, improves readability, and aligns with the AOSP Java Code Style for Contributors

Figure 6: Kotlin Warning – Redundant Conversion Method



Checklist: Kotlin Style Guide – Code Cleanliness

File/Line: BubbleGameView.kt

Issue: Redundant call of the conversion method (for example, calling `toString()` on a `String`).

Fix: Remove redundant conversion calls to simplify the code.

Why: Prevents unnecessary operations and improves code performance, following the AOSP Kotlin Style Guidelines

Katherine code review:

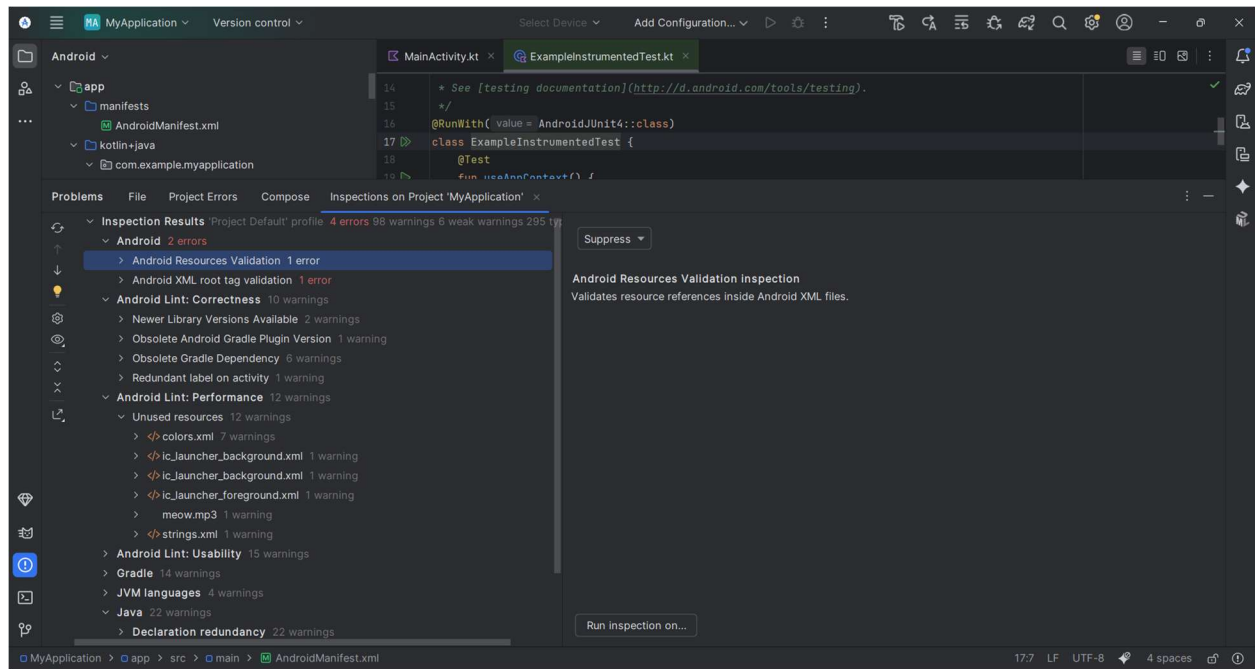
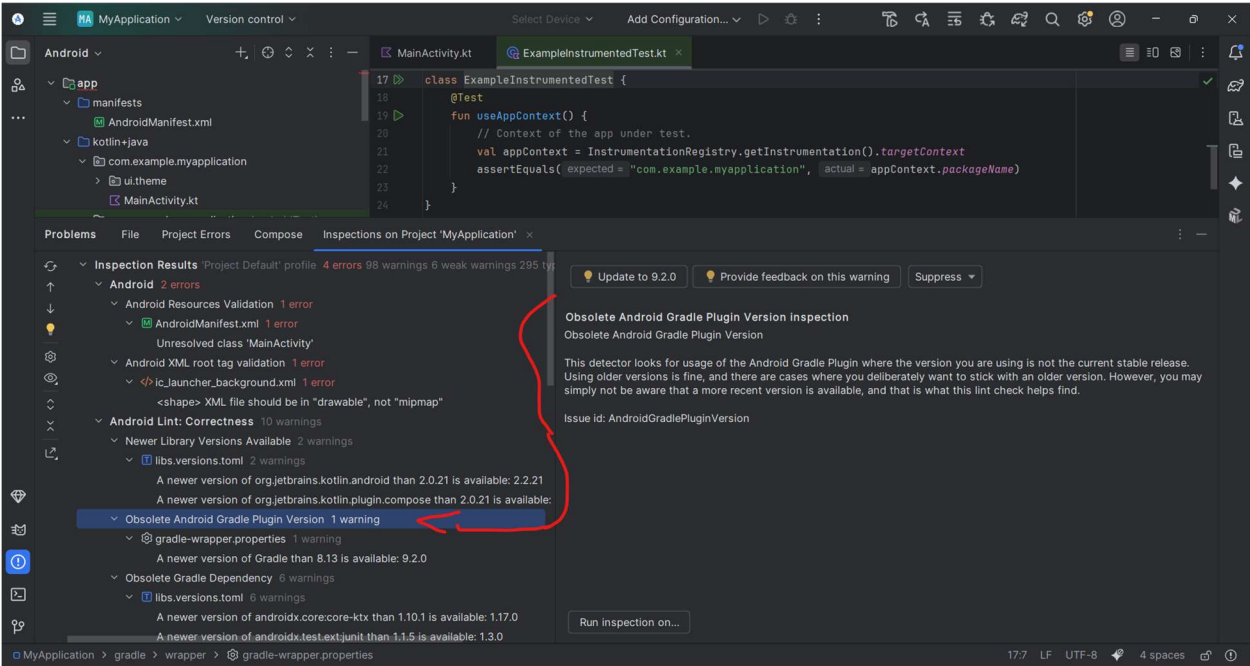
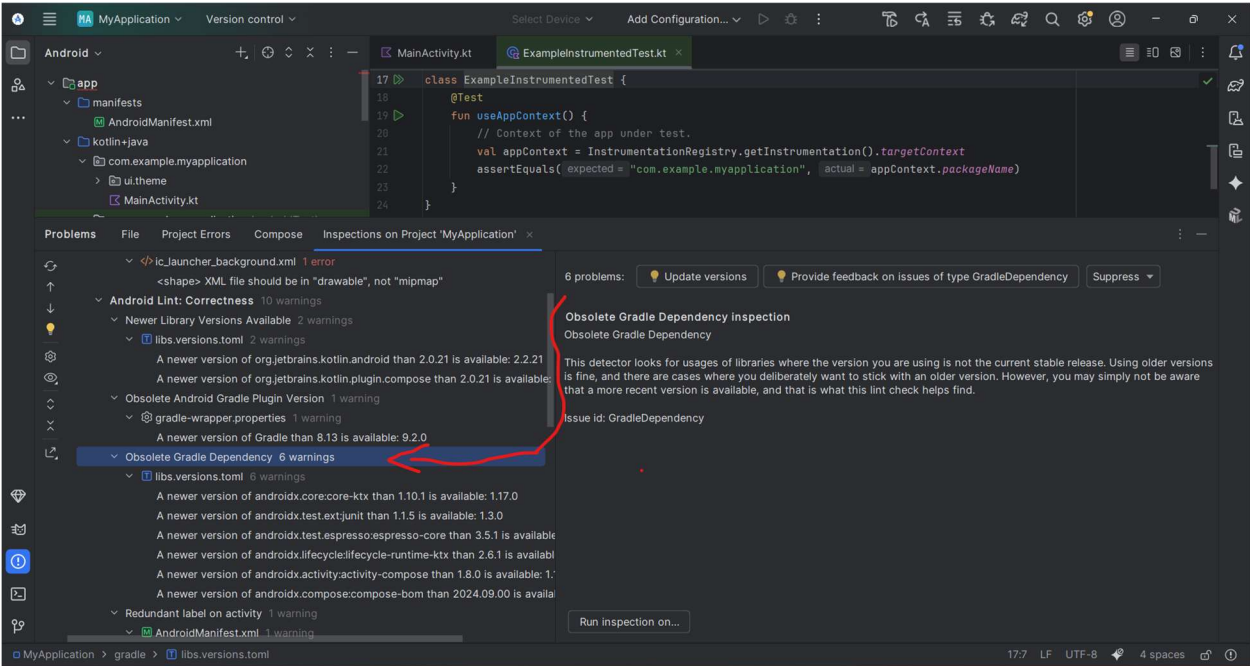


Figure 1: (Obsolete Android Gradle Plugin Version)



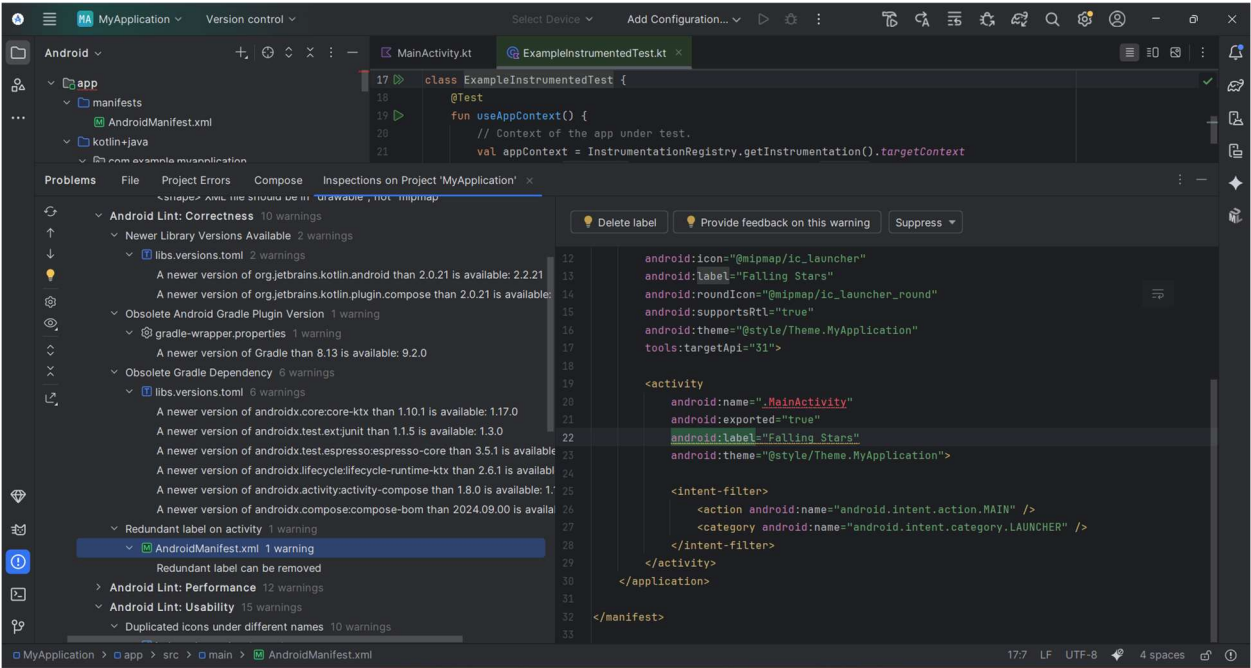
File	Checklist / Rule	Issue Found	Suggested Fix / Action	Reason / Benefit
gradle-wrapper.properties	Performance – Gradle Plugin	The project is using an old Gradle plugin version.	Update the Gradle plugin version to the latest stable release.	Improves build performance, security, and compatibility with new Android tools.

Figure 2: (Obsolete Gradle Dependency)



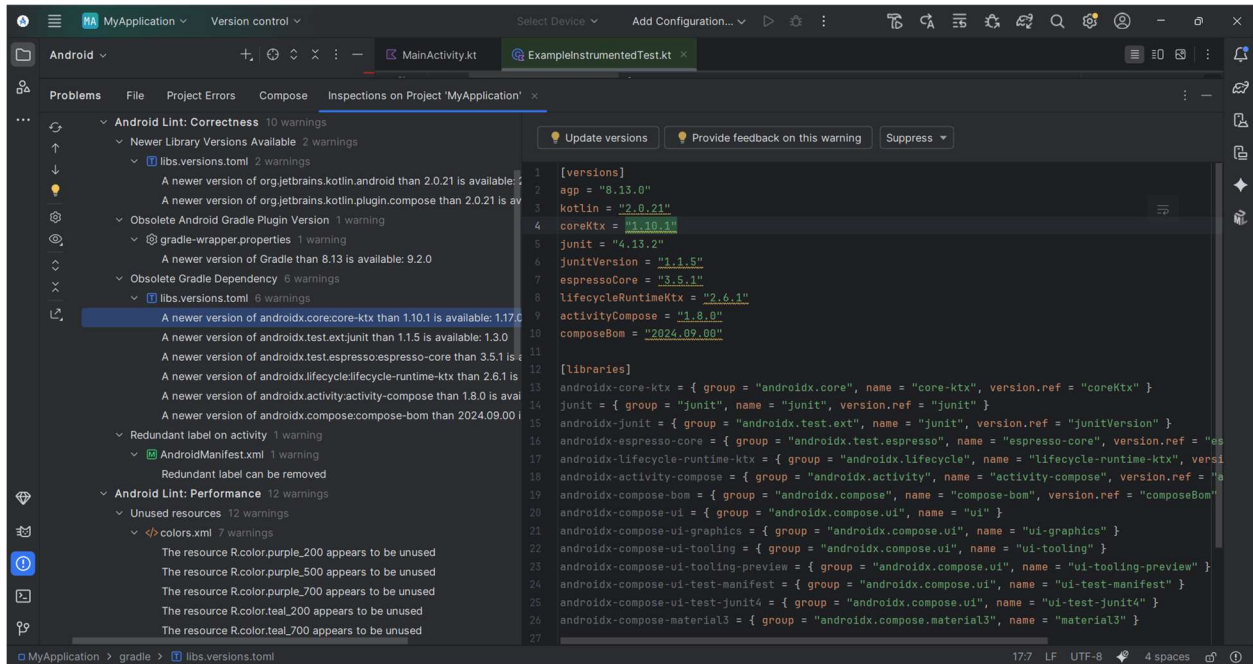
File	Checklist / Rule	Issue Found	Suggested Fix / Action	Reason / Benefit
libs.versions.toml	Performance – Gradle Dependency	Some Gradle dependencies are using old versions.	Update the Gradle dependency versions to the latest stable ones.	Keeps the project up to date and improves performance and stability.

Figure 3 (Redundant label on activity)



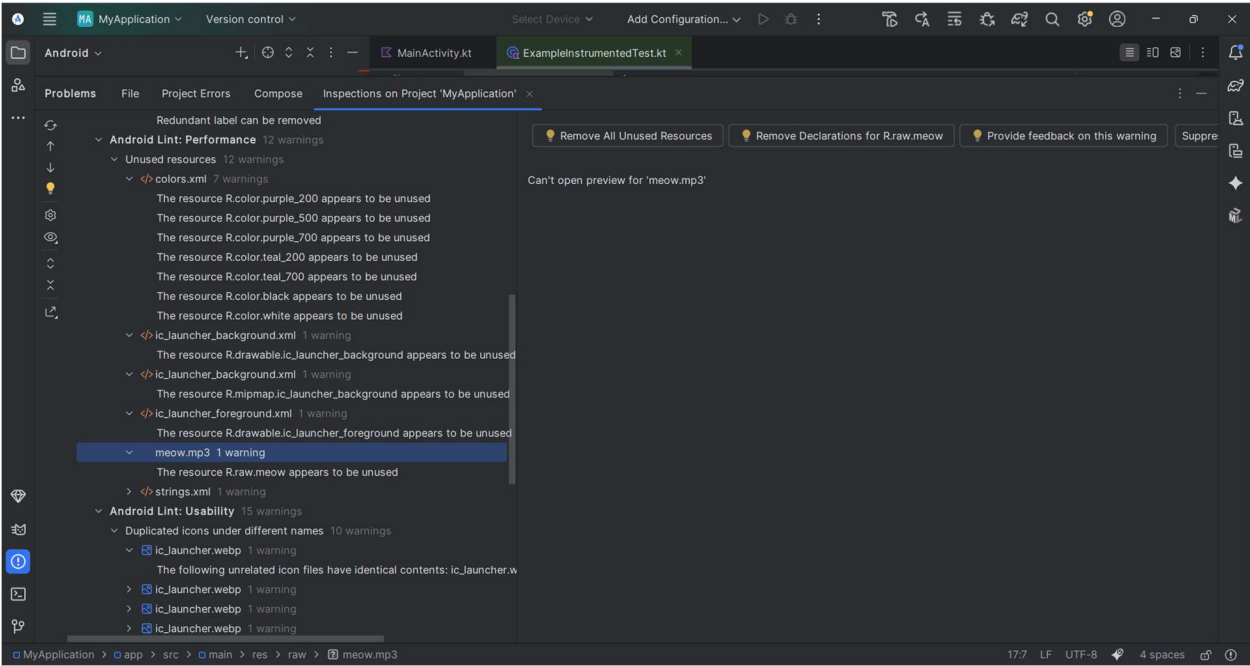
File	Checklist / Rule	Issue Found	Suggested Fix / Action	Reason / Benefit
AndroidManifest.xml	Code Optimization – Redundant Label	The activity has a redundant label.	Remove the redundant label from the activity tag.	Keeps the manifest file clean and easy to read.

Figure 4:



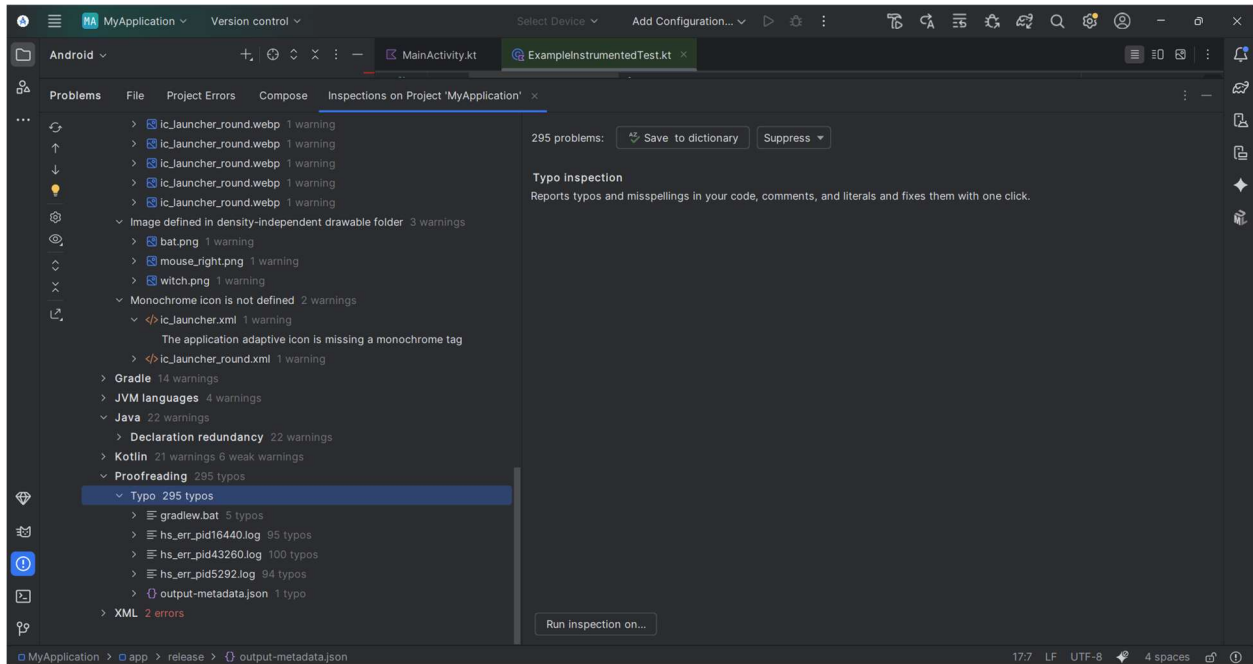
File	Checklist / Rule	Issue Found	Suggested Fix / Action	Reason / Benefit
libs.versions.toml	Dependency Management – Outdated Library Versions	Some libraries are using old versions (like Kotlin and Compose).	Update the libraries in libs.versions.toml to the latest versions.	Improves app performance, stability, and keeps the project compatible with new Android tools.

Figure 5: Performance – Unused Resources

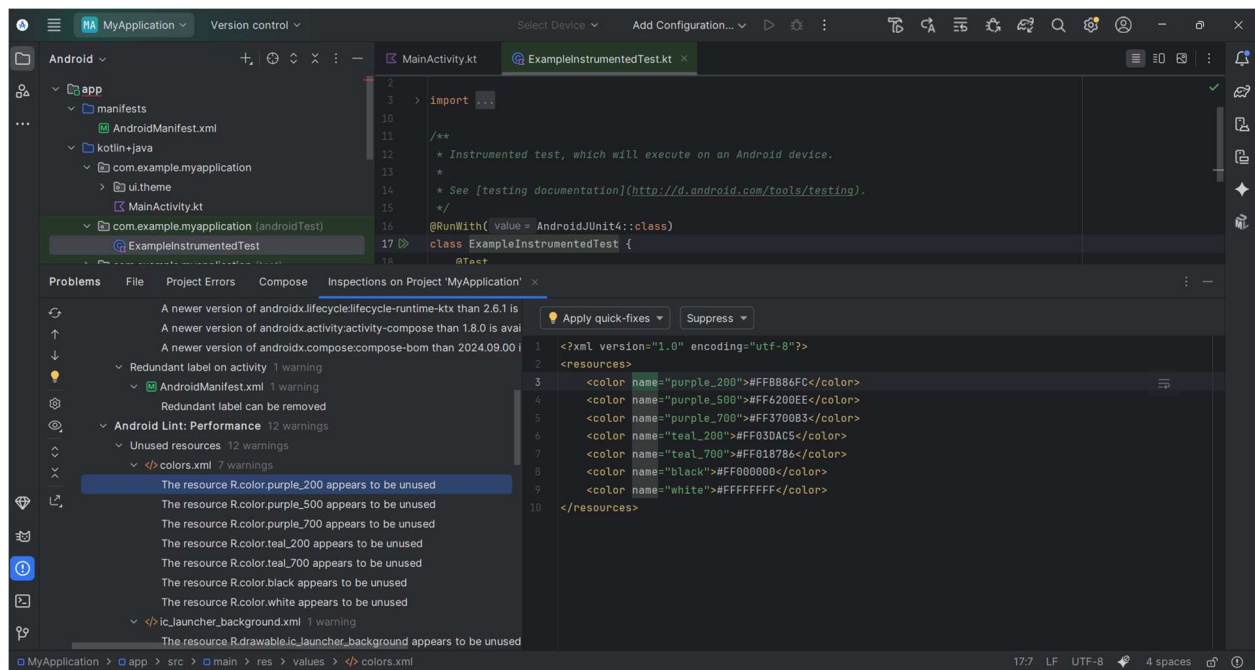


File	Checklist / Rule	Issue Found	Suggested Fix / Action	Reason / Benefit
res/raw/meow.mp3	Performance – Unused Resources	The audio file meow.mp3 is not used anywhere in the project.	Remove the unused file or reference it in code if it's required.	Reduces app size, improves build performance, and keeps the project clean.

Figure 6: Proofreading – Typos



File	Checklist / Rule	Issue Found	Suggested Fix / Action	Reason / Benefit
Multiple files	Proofreading – Typos	There are many misspelled or unrecognized words found in the code and comments.	Review and fix spelling mistakes or use the Save to dictionary option for valid technical terms.	Improves readability and makes the project more professional.



Package / File / Line	Rule # / Name	Comment (Issue Found)	Resolution (Suggested Fix)
Project root → gradle-wrapper.properties	SMELL-P01 – Obsolete Gradle Plugin	Using an old Android Gradle Plugin version.	Update to latest stable AGP (e.g., 9.2+) in gradle-wrapper.properties / Gradle settings.
Project root → libs.versions.toml	SMELL-D02 – Outdated Dependencies	Kotlin, Compose, JUnit and other libs are old.	Update versions in libs.versions.toml to current stable releases.
app/src/main/AndroidManifest.xml → <activity ...>	SMELL-R01 – Redundant Label	Duplicate android:label on activity.	Remove the extra label line; keep only one needed value.
app/src/main/AndroidManifest.xml → <application ...>	SMELL-U03 – Missing Monochrome Icon	No monochrome icon for Android 13+.	Add a monochrome icon in mipmap-anydpi-v26 and reference it, or remove the unused attribute.
app/src/main/res/raw/meow.mp3	SMELL-U01 – Unused Resource	meow.mp3 is not used anywhere.	Delete the file or use it from code if required.
app/src/main/res/mipmap-* (launcher icons)	SMELL-D01 – Duplicated Icons	Multiple launcher icons with identical content.	Keep only needed foreground/background launcher icons; remove duplicates.
app/src/main/res/values/colors.xml	SMELL-U02 – Unused Color Values	Several colors are never referenced.	Remove unused color entries.
Multiple files (comments / strings)	SMELL-T01 – Typos	Misspelled or unrecognized words.	Fix spelling or add valid tech words to dictionary.